RÉPUBLIQUE DU CAMEROUN
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MINISTÈRE DE L'ENSEIGNEMENT SUPÉRIEUR
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COMMISSION NATIONALE D'ORGANISATION DE L'EXAMEN HIGHER NATIONAL DIPLOMA EXAM (HND)
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REPUBLIC OF CAMEROON
Peace-Work-Fatherland
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MINISTRY OF HIGHER EDUCATION
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NATIONAL COMMISSION FOR THE ORGANISATION OF HIGHER NATIONAL DIPLOMA EXAM (HND) EXAM
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## National Exam of Higher National Diploma-New program – 2020 Session

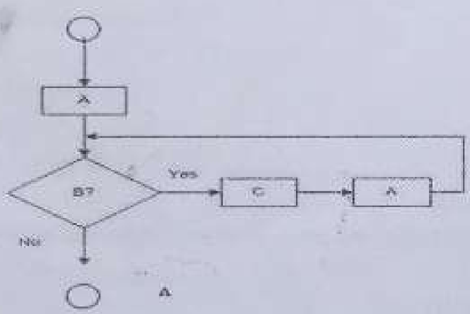
Spécialty/option : Software Engineeering
Paper : Case Study
Duration : 6 hours

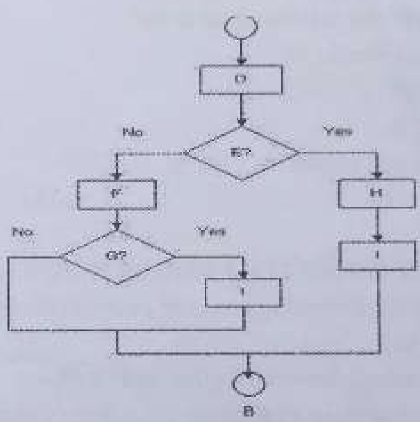
Credit : 14

### SECTION A: ALGORITHM AND PROGRAMMING. (50marks)

**I- DATA STRUCTURE AND ALGORITHMS (10 marks)**

1. Consider the flowcharts below. Write pseudocode for each example (A and B) making sure your pseudocode is structured but accomplishes the same tasks as the flowchart segment. **(2+3 marks)**




2. The statements below show some features of "Big-Oh" notation for the functions f ≡ f(n) and g ≡ g(n). Determine whether each statement is TRUE or FALSE and correct the formula in the latter case. **(5 marks)**

| Statement | Is it TRUE or FALSE? | If it is FALSE then write the correct formula |
|---|---|---|
| Rule of sums: $O(f+g)=O(f)+O(g)$ | | |
| Rule of products: $O(f\cdot g)=O(f)\cdot O(g)$ | | |
| Transitivity: if $g=O(f)$ and $h=O(f)$ then $g=O(h)$ | | |

**II. STRUCTURED PROGRAMMING (15 MARKS)**

1. Variables are very important for almost all mathematical programs in c programming. What are the rules to follow in when declaring variables in c programming? **(3 marks)**
2. List the four ways in which a function can be declared in c programming giving their respective examples. **(5marks)**

3. What is the size in byte of the array **int arr[10]**, for both a 32bit and 64bit system? **(2marks)**
4. Consider the code snippet below. Calculate the size of the structure named ***employee*** given that int is 4byte, char is 1byte, and float is 4byte. **(2 marks)**

```
struct employee
{  int id;
   char name[20];
   float salary;
};
```

5. What be the output of the following program?

```
#include <stdio.h>
int main()
{
   int a = 1, b = 1, c;
   c = a++ + b;
   printf("%d, %d", a, b);
}.
```

**(2 marks)**

6. What will the following code do?

```
#include <stdio.h>
int main() {
FILE *fp;
fp  = fopen ("data.txt", "w");
}
```

**(1 mark)**

## III. OBJECT ORIENTED PROGRAMMING (15 MARKS)

1. Define the following terms or expressions related to object oriented programming : Inheritance, Modularity **(2marks)**
2. Differentiate between a class and an object **(2marks)**
3. From what does an abstract class differs fr[illegible] concrete class? **(2marks)**
4. Name the various existing types of inheritance. **(2marks)**
5. What is a dot (.) operator in C++? **(1 mark)**
6. Considering the following C++ program:

```
#include <iostream>
using namespace std;

class Person
{
   public:
     string profession;
     int age;

     Person(): profession("unemployed"), age(16) { }
     void display()
     {
        cout << "My profession is: " << profession << endl;
        cout << "My age is: " << age << endl;
        walk();
        talk();
     }
```

```
        void walk() { cout << "I can walk." << endl; }
        void talk() { cout << "I can talk." << endl; }
};

// MathsTeacher class is derived from base class Person.
class MathsTeacher : public Person
{
    public:
        void teachMaths() { cout << "I can teach Maths." << endl; }
};

// Footballer class is derived from base class Person.
class Footballer : public Person
{
    public:
        void playFootball() { cout << "I can play Football." << endl; }
};

int main()
{
    MathsTeacher teacher;
    teacher.profession = "Teacher";
    teacher.age = 23;
    teacher.display();
    teacher.teachMaths();

    Footballer footballer;
    footballer.profession = "Footballer";
    footballer.age = 19;
    footballer.display();
    footballer.playFootball();

    return 0;
}
```

**a.** Which Object Oriented Concepts are implemented in this program ? **(2 marks)**

**b.** Analyse line by line and give the output of the program above. **(4 marks)**

## IV. OOM-UML (10MARKS)

We want to model a system for management of flights and pilots.

An airline operates flights. Each airline has an ID. Each flight has an ID a departure airport and an arrival airport: an airport as a unique identifier. Each flight has a pilot and a co-pilot, and it uses an aircraft of a certain type; a flight has also a departure time and an arrival time.

An airline owns a set of aircrafts of different types. An aircraft can be in a working state or it can be under repair. In a particular moment an aircraft can be landed or airborne. A company has a set of pilots: each pilot has an experience level: 1 is minimum, 3 is maximum.

A type of aero plane may need a particular number of pilots, with a different role (e.g.: captain, co-pilot, navigator): there must be at least one captain and one co-pilot, and a captain must have a level 3.

a. Differentiate between static and dynamic UML diagrams with two examples in each case. (4marks)

b. Identify all the classes and the relationship between them. **(3marks)**

c. Produce the Class diagram of the system. **(3marks)**

## <u>SECTION B:</u> DATABASE DEVELOPMENT AND ADMINISTRATION (20 MARKS)

**<u>EXERCISE IV</u> (20 marks)**

1. Think of an organizational database in which some of the fields in the CUSTOMER table must have the given data types. Explain what they mean and how they are used:
   i) Customer ID (auto numeric field)
   ii) Customer Name (text field)
   iii) Fee Paid (decimal field)
   iv) Pay Date (date field). **(2 x 4 = 8 marks)**
2. Table 1 contains sample data for vehicles and for operators who ply these vehicles. In discussing these data with users, we find that vehicle ID (but not descriptions) uniquely identify vehicles and that operator names uniquely identify operators.
   a) Convert this table to a relation (named VEHICLE OPERATOR) in first normal form. Illustrate the relation with the sample data in the table. **(2 marks)**
   b) List the functional dependencies in VEHICLE OPERATOR and identify a candidate key. **(3 marks)**
   c) For the relation VEHICLE OPERATOR, identify each of the following: an insert anomaly, a delete anomaly, and a modification anomaly. **(3 marks)**
   d) Draw a relational schema for VEHICLE OPERATOR and show the functional dependencies. **(3 marks)**
   e) In what normal form is this relation? **(1 mark)**

**Table 1:** Sample Data for Vehicles and Operations

| VehicleID | Description | Operator | Route | Tarrif Per Mile |
|---|---|---|---|---|
| V1 | Luxury | Polax | Grand Trail | 100 |
| | | Ubet | East Route | 150 |
| V2 | Comfort | Polax | Grand Trail | 45 |
| | | Ubet | East Route | 60 |
| | | Minim | South Trunk | 35 |

## <u>SECTION C :</u> WEB DESIGN

**<u>Exercise 6</u>:HTML, JS, CSS, PHP(15 marks)**

1. What is JavaScript used for? **(1 mark)**
2. Is it possible to implement a web project without using JavaScript? **(1 mark)**
3. The purpose of building a web application is to make it accessible to the whole world, explain the process of deploying a web application built locally. **(2 marks)**
4. Define PHP, what are the various tools needed to develop PHP project? **(1,2 marks)**
5. Create a form (HTML+ CSS) to authenticate users (Login, Password) *see figure 1* **(4marks)**
6. Creates a php file (process.php) to get the parameters from the form and display them. **(4 marks)**

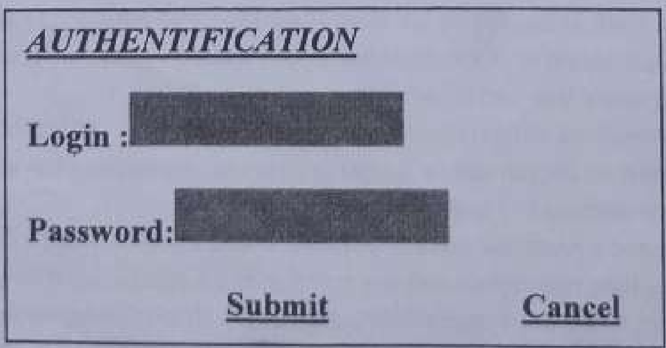



***Figure 1: login form***

## SECTION D: NETWORKING (15 MARKS)

**EXERCISE VI (15 marks)**

You are required to setup a LAN in your school secretariat to serve both wired and wireless users, and to connect to the Internet.

The network will consist of 10 PCs, one printer accessible by all the PCs, 7 laptops.

a) Name the other network devices you would need in order to setup the LAN. **(3 marks)**

b) Name the media type required, and connectors required. **(2 marks)**

c) What would you need to install in order to protect your network from external attack? **(1 mark)**

d) Upon successfully setting up the LAN, you still don't have internet connection. Who should you contact for internet connection? **(1 mark)**

e) List two different connection options that can provided to you by d) above. **(1 mark)**

f) Given the network address 192.168.1.0,

   i) How many usable host addresses does it provide? **(1 mark)**

   ii) Which address would most likely be assigned to the device interface that connects the LAN to the Internet? **(1 mark)**

   iii) State the address that a remote host will use to communicate with all hosts in this LAN. **(1 mark)**

g) Sketch a physical topology of your network specifying the different cable types used in connecting the different devices. **(4 marks)**